**What is a psychological assessment?**

Psychological assessment is a series of tests conducted by a psychologist, to gather information about how people think, feel, behave and react. The findings are used to develop a report of the person's abilities and behavior—known as a psychological report—which is then used as a basis to make recommendations for the individual's treatment.
Psychological assessments and reports are used in other fields as well—like in the case of career planning for young adults or in the job application process to determine how well an applicant will fit into the position.

The procedures used to create an assessment are:

- Interviews
- Observation
- Written assessment
- Consultation with other mental health professionals
- Formal psychological tests

**What is a psychological test?**

A psychological test is used to measure an individual's different abilities, such as their aptitude in a particular field, cognitive functions like memory and intelligence or even traits e.g. introvert and extrovert etc. These tests are based on scientifically tested psychological theories.
The format of a test can vary from pencil and paper tasks to computer-based ones. They include activities such as puzzle-solving, drawing, logic problem solving, and memory games.

Some tests also use techniques—known as projective techniques—which aim to access the unconscious. In these instances, the subject's responses are analyzed through psychological interpretation and more complex procedures than the non-projective techniques mentioned above. For example, the Rorschach test, popularly known as the ink-blot test can provide insight into the person's personality and emotional functioning.

Psychological tests may also involve observing someone's interactions and behavior. Based on the result of the test, conclusion will be drawn about the individual's abilities and potential.

**Difference between test and assessment**

- A test is a measurement device or technique used to quantify behavior or aid in the understanding and prediction of behavior"
- Assessment includes more than just tests. A typical assessment plan may include a test or a battery of tests, interview, behavioral observation, and case history data

**Uses of Psychological Test**

1. **Understanding People**: They help understand how people think, feel, or behave.

2. **Comparing Differences:** These tests compare how individuals are different from each other.

3. **Measuring Changes:** They measure how someone might change over time in their thoughts or feelings.

4. **Assessing Abilities:** They check what someone is good at or what they might struggle with, like in school or work.

5. **Diagnosing Problems:** Psychological tests help identify issues like learning disabilities, emotional struggles, or mental health conditions.

6. **Guiding Decisions:** They assist professionals, like psychologists or counselors, in making decisions about treatment or support for someone.

7. **Assessment of Skills and Abilities:** Evaluating strengths and weaknesses in various areas, such as learning, memory, or problem-solving.

8. **Identifying Emotional Health:** Assessing emotions and mental health conditions like anxiety, depression, or stress.

9. **Measuring Personality Traits:** Analyzing different personality aspects such as introversion/extroversion, conscientiousness, or openness to experience.

10. **Diagnosing Disorders and Conditions:** Detecting conditions like ADHD, autism, or specific learning disabilities.

11. **Research and Development:** Contributing to research by providing data and insights into human behavior, cognition, and emotions for developing new theories or treatments.

**Types of Psychological tests**



1. Individual versus group tests
2. Ability, achievement, or aptitude tests
3. Intelligence versus personality tests
4. Speed test versus ability tests
5. Structured personality tests versus projective tests
6. Verbal versus non-verbal/performance tests.
7. Commercial copyrighted tests versus available to all test

**Individual versus Group Tests:**

Some tests are for one person at a time (like WAIS, WISC), while others can be done with many people together (like Raven's Progressive Matrices). Individual tests need one examiner for one person, while group tests are quicker, with multiple people and often use multiple-choice questions. Group tests are easier to score and can be administered through recorded instructions or computers. However, they might not be ideal when a good connection between the person being tested and the examiner is necessary.

**Ability, achievement, or aptitude tests**

Ability tests, like intelligence tests, measure your thinking and learning skills, providing an IQ score. They show what you know at a certain time and your level of skills in different areas.

Achievement tests check what you've learned from specific training or programs, like SATs at the end of courses, to see if you met the goals.

Aptitude tests measure the overall effect of your life experiences on your learning abilities. They predict how well you might do in the future, unrelated to specific training. Differential Aptitude Tests (DAT) are common examples.

**Intelligence Verses Personality Tests:**

Even though the test names suggest what they measure, it's important to know that personality tests don't tell you about IQ, and IQ tests can't tell you about personality traits. Many confuse these, but they're different and come in many varieties, both being widely available**.**

**Speed Tests versus Ability Tests:**

In ability tests, like IQ tests, the focus is on measuring the level of ability, not how fast you finish. Some tests, such as Raven’s Progressive Matrices, don’t have a time limit. However, in other tests like Clerical Speed and Accuracy Test, how quickly you perform is important. Power tests have a time limit, but it's long enough for everyone to finish.

**Structured Personality Tests versus Projective Tests:**

Personality tests come in different types and measure usual behavior like traits. There are two main categories: structured tests and projective tests. Structured tests have fixed answer options, like true/false or multiple-choice questions, where you pick the one that best describes you. These tests are often self-report type.

Projective tests, unlike structured ones, are harder to score and interpret. They involve showing vague images or words to people who then describe what they see or complete the task. For example, Rorschach’s Inkblots or TAT, where people tell stories about pictures, are used to understand personality. Other tests, like HTP or RISB, involve drawing or completing sentences to reveal traits or tendencies.

**Verbal versus Nonverbal/ Performance Tests:**

Many tests rely on language skills, especially IQ tests and structured personality tests. However, tests like Raven’s Progressive Matrices don’t measure verbal abilities or use language.

**Commercial-Copyrighted Tests Versus “Available To All”/Online Tests:**

Most copyrighted IQ tests, like WAIS or WISC, can't be copied or used without permission as it's unethical and diminishes their value if people already know the content. However, some tests available online or in books are mainly for research, not for diagnosing or screening. Personality tests are more commonly found online, like the Multidimensional Health Locus of Control Scale or Self Efficacy Scale, which authors have made available for use, downloading, and printing, offering valuable research evidence**.**

**Ethical, Legal and Professional issues in assessment**

1. **Confidentiality and Privacy:** Safeguarding sensitive information collected during assessments to ensure it remains private and is shared only with authorized individuals or for authorized purposes.

2. **Informed Consent:** Obtaining explicit consent from participants or individuals being assessed, providing comprehensive information about the assessment process, its objectives, and how the results will be used.

3. **Fairness and Equity:** Ensuring assessments are conducted impartially, free from bias or discrimination based on factors like race, gender, ethnicity, or socioeconomic status.

4. **Validity and Reliability:** Ensuring assessment tools accurately measure what they are intended to measure and produce consistent, dependable results.

5. **Professional Competence:** Conducting assessments within one's area of expertise, adhering to professional standards, and maintaining ongoing professional development.

6. **Legal Compliance:** Abiding by laws and regulations governing assessments, such as those related to data protection, disability accommodations, and informed consent.

7. **Avoiding Conflicts of Interest:** Preventing situations where personal interests or biases could influence assessment outcomes, ensuring objectivity and fairness.

8. **Appropriate Use of Results:** Using assessment outcomes responsibly, ensuring accurate interpretation, and avoiding misuse that could have adverse consequences for individuals.

9. **Cultural Sensitivity:** Recognizing and addressing cultural influences that might affect the assessment process or interpretation of results, ensuring assessments are culturally relevant and unbiased.

10. **Clear Communication and Feedback:** Providing understandable and meaningful feedback to individuals being assessed, ensuring they comprehend the results and their implications for further action or support.

Adherence to these ethical, legal, and professional principles is vital in maintaining integrity, fairness, and respect for individuals throughout the assessment process.

**Characteristics of a Test**

- **Reliability**

**What Is Reliability in Psychology?**

**According to Kaplan & Saccuzzo (2001) reliability is "the extent to which a score or measure is free from measurement error. Theoretically, reliability is the ratio of true score variance to observed score variance."**

**Measurement Error**

The classical test score theory implies that everyone can obtain a true score on any test or measure if the measure is free of error. But we know that there is perhaps no measure that can be considered as totally error free. There is always some chance of error. This further implies that the scores that we obtain on different measures are not the true scores of the test taker. These are the observed scores plus error. In other words the observed score is not the true score of a test taker, implying that the observed score may not be a 100% true representative of a person's traits or abilities. Here we need to realize that the term 'error' is not used to indicate that something has gone wrong or we have made some 'mistake'. Error, in this context, refers to the amount and extent of variance that may be expected in results. The observed score is therefore the sum of true score and error i.e.

$X= T+E$

Here X refers to the observed score, T is the true score, and E is the error that can be expected in the measurement

**Sources of Error in Test Scores:** Psychologists and educationists try to estimate the amount and degree of error that may be expected in their measurement. That is the main reason why test developers and test administrators emphasize on uniform testing procedures besides controlling other possible sources of error.

The following are a few of the variables that may be possibly causing error in measurement:

A. Test Related Factors:
    - Difficulty level (too difficult or too easy)
    - Length of the test (too long causing fatigue or boredom)
    - Domain or content (not suitable for all test takers…..suitable for some but not all)
    - Items may not be representing the domain or content
    - Time limit (may cause stress or a handicap)

B. Test Administration Process:
    - Poor and not uniform testing conditions (physical setting and environment)
    - Poor, faulty, improperly worded, improperly delivered, not uniform instructions

- Test administrator’s personality ( different administrators in different situations having different personality styles)
- Rapport ( poor, too much, or too little)

C. Examinee Related Variables:
- Prior learning and experience
- Individual differences (within group differences; Personality styles, stress tolerance level, IQ level, emotionality, motivation, knowledge)
- Difference from the normative sample ( no group is identical to the normative sample; every group is different from every other group)
- Within- person differences (the same persons may change over time (life, academic, and professional experiences; physical conditions, health, motivational level, emotional state etc.)

**Correlational and reliability**

Correlation shows the relation between 2 variables, and its measure by a statistics called the correlation of coefficient.

- Coefficient of correlation tells us two things about a relationship; the magnitude of relationship and the direction of relationship. Magnitude means the size of correlation whereas direction indicates whether the correlation is positive or negative.
- The size of correlation range between -1 (perfect negative) and +1(perfect positive ) with a zero (no correlation at all) value in between
- The value of coefficient of a correlation is always 1 or less than 1
- The closer is a value of correlation to one, the stronger is the correlation e.g. r= 0.9.
- The closer a coefficient of correlation is to zero, the weaker will be the relationship.

**Types of Reliability**

According to the definition given by Anastasi & Urbina (2007) reliability refers to “the consistency of the scores obtained by the same persons when they are reexamined with the same test on different occasions or with different sets of equivalent items, or under other variable examining conditions”. There are two main types of reliability: internal reliability and external reliability.

1. **Internal reliability** means that the measure has consistency within itself. In other words, the same question posed differently would produce the same results. It is often measured using the split-half method.

2. **External reliability** refers to how results compare to results between individuals and across time. It is often measured using test-retest, inter-rater, and parallel forms methods.

## Measurement of Reliability

### Test-Retest Reliability

Test-retest reliability is a measure of the consistency of a psychological test or assessment. This kind of reliability is used to determine the consistency of a test across time. Test-retest reliability is best used for things that are stable over time, such as intelligence.

Test-retest reliability is measured by administering a test twice at two different points in time. This type of reliability assumes that there will be no change in the quality or construct being measured. In most cases, reliability will be higher when little time has passed between tests.

The test-retest method is just one of the ways that can be used to determine the reliability of a measurement. Other techniques that can be used include inter-rater reliability, internal consistency, and parallel-forms reliability.

A major **advantage** of this method is that there is maximum control over the test taker and test item variables. The subjects are the same and the test items are also the same.

### Drawbacks

1. **Changing Conditions**: Sometimes, the environment or circumstances might change between two tests. Things like noise or temperature can affect how well someone performs.
2. **Changes in People:** Test takers might change too. Their feelings, motivation, social situations, or even their health might be different the second time they take the test.
3. **Practice Effect:** Doing the same test twice might make it less challenging or more familiar for the person. Or it might become boring. This can affect their performance.
4. **Learning from the Test:** Some tests are such that people might remember the questions or figure out how to answer them better the second time. This can especially affect tests that measure thinking skills like problem-solving, reasoning, or math.
5. **Time Gap Matters**: The time between the two tests is important. If the gap is short, the results might be more similar. But if it's longer, the results might differ more. Test makers should mention the time gap when they talk about how reliable the test

Test- retest reliability is denoted as **rtt.**

### Inter-Rater Reliability

This type of reliability is assessed by having two or more independent judges score the test. The scores are then compared to determine the consistency of the examiners estimates.

One way to test inter-rater reliability is to have each examiner assign each test item a score. For example, each examiner might score items on a scale from 1 to 10. Next, you would calculate the correlation between the two ratings to determine the level of inter-rater reliability.

Another means of testing inter-rater reliability is to have examiners determine which category each observation falls into and then calculate the percentage of agreement between the examiners. So, if the examiners agree 8 out of 10 times, the test has an 80% inter-rater reliability rate.

**Parallel-Forms Reliability / Alternate-Form**

Parallel-forms reliability is assessed by comparing two different tests that were created using the same content. This is accomplished by creating a large pool of test items that measure the same quality and then randomly dividing the items into two separate tests. The two tests should then be administered to the same subjects at the same time.

Independently developed and completely parallel or equivalent forms of the same measure. They should match each other in all respects including:

- Same specifications
- Same instructions
- Same time limit
- Same content
- Same number of items
- Same item format
- Difficulty level

Consistently and stability across the parallel of forms

Consistency over time

| | Form A | Form B |
|---|---|---|
| Occasion 1 | Group I (50% of total) | Group 2 (50% of total) |
| | Interval | |
| Occasion 2 | Group 2 (50% of total) | Group I (50% of total) |

**Advantages**

1. Increases Confidence: Using two different tests boosts confidence in the accuracy of results.

2. Minimizes Memory Effects: Prevents people from recalling answers between tests, ensuring more reliable results.
3. Provides Diverse Insights: Different tests offer a broader understanding of someone's abilities.
4. Identifies Test Issues: Consistent results indicate a reliable test, while differences might signal problems.
5. Promotes Fairness: Using different forms ensures fairness without favoritism for specific questions or formats.
6. Enhances Flexibility: Having various test versions can be advantageous in different situations.

**Drawbacks**

- Hard to Create: Making two tests that are equally good at measuring the same thing is tough. They need to be equally challenging and cover the same topics.
- Difficult to Manage: Giving two different tests to the same group can be tough to organize, especially in big groups or when testing conditions need to be the same.
- Small Differences: Even if we try hard, there might be tiny differences between the two tests that affect how well they measure the same thing.
- Not Always Possible: Some things are hard to test with two different versions. For example, certain skills or abstract ideas might not easily work with different test forms.
- Practice Effect: If people get used to the type of questions on the first test, they might do better on the second test just because they're more familiar with it. This can mess up the reliability of the test.
- Takes Time and Money: Creating and checking two versions of a test needs a lot of time, effort, and money. This might not be possible or practical in some situations.

**Split half Method Reliability**

This form of reliability is used to judge the consistency of results across items on the same test.

Essentially, you are comparing test items that measure the same construct to determine the tests internal consistency. It is often referred to as the split-half method of measuring reliability.

When you see a question that seems very similar to another test question, it may indicate that the two questions are being used to assess reliability.

Because the two questions are similar and designed to measure the same thing, the test taker should answer both questions the same, which would indicate that the test has internal consistency.

The test can be divided in many ways:

- Divide test into equal number of items.

The first 50% items are considered as one half and the remaining 50% as the other half. Scores are calculated separately for the two halves. There is one limitation of dividing the test this way. There is likelihood that the two halves may be having different difficulty levels. The initial items may be easier than the items in the second half. Also, items in the two halves formed in this way may pertain to different content areas

- A better option can be to consider odd numbered items in one half, and the even numbered items in the other half.

This formula makes it possible to estimate the internal consistency reliability of the test from the reliability of the two halves of the test. The formula is used for the computation of reliability while taking care of the fact that a test has been shortened or lengthened. The formula also considers the number of times the test has been shortened or lengthened.

$$r_{nn}=\frac{nr_{tt}}{1+(n-1)r_{tt}}$$

Spearman- Brown formula is given below:

rnn= estimated coefficient

rtt= obtained coefficient

n= number of times the test has been lengthened or shortened e.g. two times, three times etc

In split- half reliability the test is reduced to half, therefore the reliability of the full test is calculated using the Spearman- Brown formula that involves doubling the length of the test, as follows:

$$r_{tt}=\frac{2r_{hh}}{1+2r_{hh}}$$

**Draw backs**

1. **Splitting Impact: Reliability** varies based on how the test is split**.**
2. **No Item-Level Assessment:** Doesn't gauge consistency within individual questions.
3. **Small Group Challenges**: Less accurate with fewer test-takers.
4. **Accuracy Risks:** Short tests may give unreliable reliability estimates**.**
5. **Assumption of Equal Difficulty:** Assumes both halves are equally challenging.
6. **Chance Influence:** Splits may sometimes result in random outcomes.
7. **Limited for Diverse Tests:** Not ideal for tests with varied question types**.**
8. **Length Influence:** Longer tests tend to offer more reliable estimates.
9. **Mixed Subject Disadvantage:** Might not accurately reflect overall consistency in tests covering different subjects.

**Kuder- Richardson Reliability:**

The Kuder-Richardson method is somewhat like the split-half method because both check how reliable a test is using just one administration. But, they're different in how they look at consistency. Instead of splitting the test into halves, the Kuder-Richardson method looks at how each question performs. Basically, it's the average of all the split-half reliability scores you can get from a test.

$$r_{tt}=\left(\frac{n}{n-1}\right)\frac{SD_t^2-\sum pq}{SD_t^2}$$

$r_{tt}$= Coefficient of reliability of whole test
**n**= the number of items in the test
$SD_t^2$= Standard deviation of total scores on the test is $SD_t$, squared (=variance)
**p**= proportion of persons who pass each item
**q**= proportion of persons who do not pass each item
pq= product of (p) and (q) for each item
$\Sigma_{pq}$= The sum of products of (p) and (q) for all item

- Only used for that items that their answers are either right or wrong / zero or one
- Cannot use with Likert scale and having weightage

**Coefficient Alpha:**

Coefficient alpha is another method, much like Kuder-Richardson, used to check how consistent a test is. But while Kuder-Richardson works only for tests where answers are either right or wrong (like true/false), coefficient alpha works for tests where answers get different scores, like in personality tests where answers have different weights. This formula helps measure the consistency of these types of tests.

$$r_{tt} = \left(\frac{n}{n-1}\right)\frac{SD_t^2 - \sum\left(SD_i^2\right)}{SD_t^2}$$

$r_{tt}$= Coefficient of reliability of whole test
$SD_t^2$= Standard deviation of total scores on the test is $SD_t$, squared (=variance)
$SD_i^2$ = Variances for item scores
$\sum(SD_i^2)$= sum of the variances of item scores

**Factors That Can Impact Reliability**

There are a number of different factors that can have an influence on the reliability of a measure. First and perhaps most obviously, it is important that the thing that is being measured be fairly stable and consistent.

If the thing being measured is something that changes regularly, the results of the test will not be consistent.

Aspects of the testing situation can also affect reliability. For example, if the test is administered in a room that is extremely hot, respondents might be distracted and unable to complete the test to the best of their ability. This can have an influence on the reliability of the measure.

Other things like fatigue, stress, sickness, motivation, poor instructions and environmental distractions can also hurt reliability.

**Standard error of measurement**

The standard error of measurement (SEm) estimates how repeated measures of a person on the same instrument tend to be distributed around his or her “true” score. The true score is always an unknown because no measure can be constructed that provides a perfect reflection of the true score.

The Standard Error of Measurement (SEM) is a measure of how much mistakes or variability exists in a test. It's like a measure of the "ups and downs" in test scores for the same person taking the test multiple times.

To figure out SEM, you use a formula. If you know the reliability of the test (how consistent it is), and the standard deviation of the test scores, you can calculate SEM using a simple formula. The formula uses the test's reliability and the variation in scores to estimate the amount of error or variability in the test results.

$$SEM = SD_t\sqrt{1 - r_{tt}}$$

**SDt = Standard deviation of the test score**

**Rtt = reliability coefficient**

**For example**, if a test had a standard deviation of 12 and the reliability coefficient is .90, then SEM will be= 12√ 1- .9= 3.79 .The SEM is the standard deviation of the hypothetical distribution. If XYZ attains an average IQ 0f 90 in the hypothetical situation it will be her true score that may range between 90- 3.79 and 90+ 3.79.we remember that in normal distribution about 68% people fall between -1 and +1 standard deviations from the men. Therefore applying the same concept we can say that 68% people may be expected to score within this range of scores.

## Validity

Validity is the extent to which a test measures what it claims to measure. It is vital for a test to be valid in order for the results to be accurately applied and interpreted.

**Internal validity** is a measure of how well a study is conducted (its structure) and how accurately its results reflect the studied group.

**External validity** relates to how applicable the findings are in the real world. These two concepts help researchers gauge if the results of a research study are trustworthy and meaningful.

**Example:**

**Internal Validity** is like making sure that your test or experiment measures what it's supposed to measure.

Let's say you want to find out if studying with music helps students remember information better. You conduct an experiment where one group studies with music, and the other studies in silence. If the group that studied with music performs better on a test, you want to be sure it's because of the music and not because of any other factor. Internal validity involves setting up your experiment in a way that ensures the only difference between the two groups is the music while controlling other factors like study time, difficulty of the material, and individual differences.

**External Validity** is like checking if the results of your experiment can be applied to a larger group or real-life situations.

Continuing the music and studying example, let's say your experiment shows that studying with music helps students remember better. To check external validity, you might wonder if this applies to all students, not just the ones in your experiment. Do students from different schools or countries show similar results? Does this work for all subjects? Ensuring external validity means considering if your findings can be relevant and useful beyond the specific conditions of your experiment.

So, internal validity is about making sure your experiment gives accurate results, while external validity is about how broadly you can apply those results to other situations or people.

## Types of validity

There are four type of validity

1. **Construct validity** ensures the test measures what it claims to measure.
2. **Content validity** ensures the test covers all the important areas it's supposed to cover.

3. **Face validity** is about whether the test seems to be right for its purpose at first glance.
4. **Criterion validity** checks if the test results accurately predict or correspond to real outcomes or standards.

- **Construct validity**

Construct validity refers to how well a test or measurement accurately assesses the specific idea or concept it's intended to measure. It's about ensuring that the test is actually tapping into the theoretical construct or idea that it claims to evaluate.

**For example**, if you create a test to measure intelligence, construct validity would involve demonstrating that the test questions effectively capture different aspects of intelligence, such as problem-solving, reasoning, memory, and linguistic abilities. It's not just about asking questions randomly; it's about designing questions that truly represent and measure the underlying concept of intelligence.

In simple terms, construct validity is all about making sure that a test is accurately measuring the specific idea or concept it claims to measure.

- **Content validity**

Content validity refers to how well a test or assessment instrument covers all the important areas or aspects of the subject or skill it's intended to measure. It assesses whether the content of the test is an adequate and representative sample of the entire domain it aims to evaluate.

In simpler terms, content validity asks whether the test questions or items sufficiently represent the breadth and depth of the subject being tested.

**For example**, if you're creating a history exam for high school students to evaluate their knowledge of World War II, content validity would mean ensuring that the test includes questions about various critical events, key figures, causes, consequences, and different perspectives related to that historical period. Content validity ensures that the test adequately represents the entire scope of the topic it intends to assess.

Establishing content validity involves expert sjudgment, where subject matter experts review the test items to confirm that they cover the relevant content comprehensively and fairly represent the subject being measured.

In essence, content validity ensures that the test or assessment tool accurately measures the width and depth of the subject it's supposed to evaluate.

- **Face validity**